

Topological Data Analysis

Lecture 17

Mapper

Oleg Kachan

Partition



Given a set X , a clustering can be posed as a partition of X , that is

$$\mathcal{P} = \{C_1, \dots, C_n\}$$

such that

$$X = \bigcup_{i=1}^n C_i \quad C_i \cap C_j = \emptyset, \forall i, j$$

Cover



Given a set X , a cover of X is the family of subsets of X $\{C_i\}_{i \in I}$

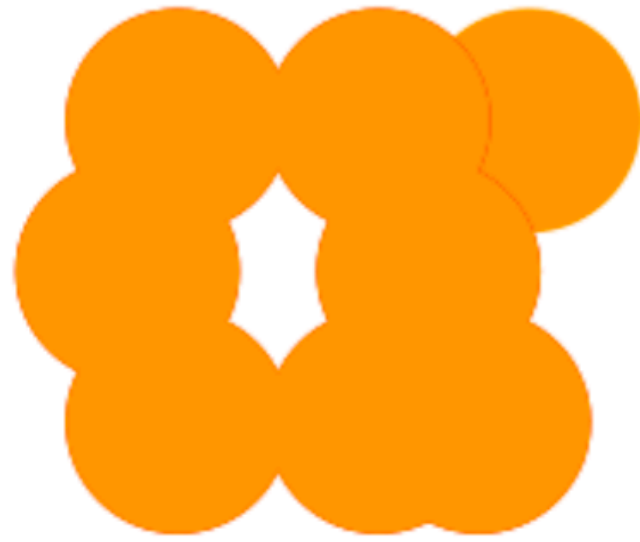
$$\mathcal{C} = \{C_1, \dots, C_n\}$$

such that

$$X = \bigcup_{i=1}^n C_i$$

Nerve of a cover

X



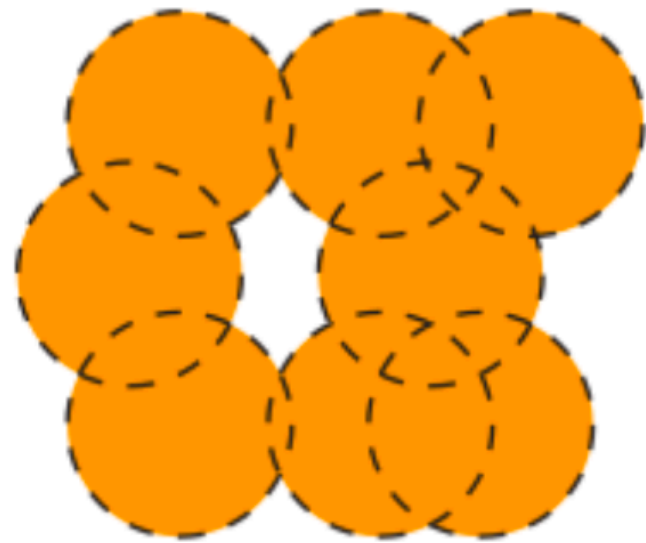
Given a family of sets $\{C_i\}_{i \in I}$, with the index set I , the nerve complex $N(C)$ is a simplicial complex with a vertex set I

$$N(C) = \left\{ J \subseteq I \mid \bigcap_{j \in J} C_j \neq \emptyset \right\}$$

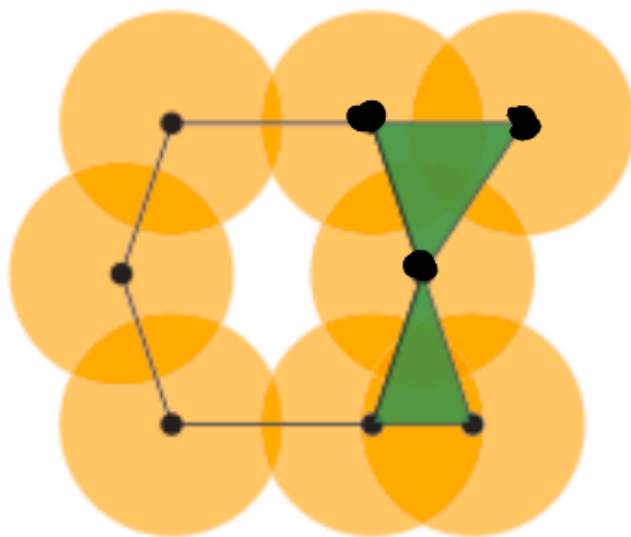
that is, a k -simplex $\sigma^{(k)}$ belonging to the nerve complex iff

$$\sigma^{(k)}\{i_0, i_1, \dots, i_k\} \in N(C) \iff C_{i_0} \cap C_{i_1} \cap \dots \cap C_{i_k} \neq \emptyset$$

$\mathcal{C}(X)$

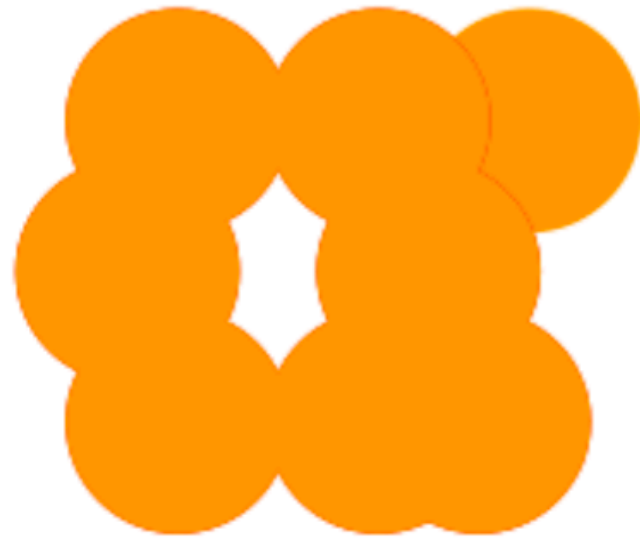


$N(\mathcal{C})$



Nerve of a cover

X



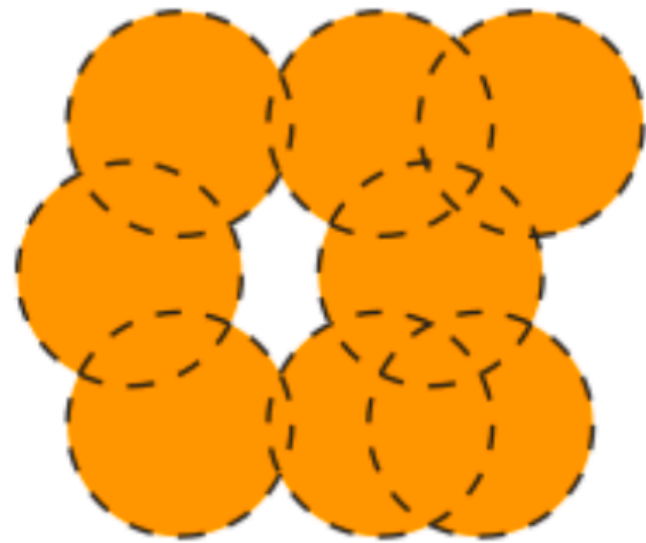
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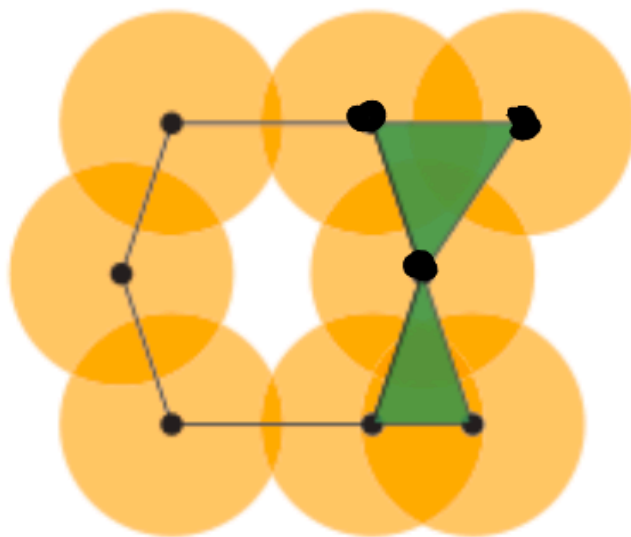


Nerve theorem

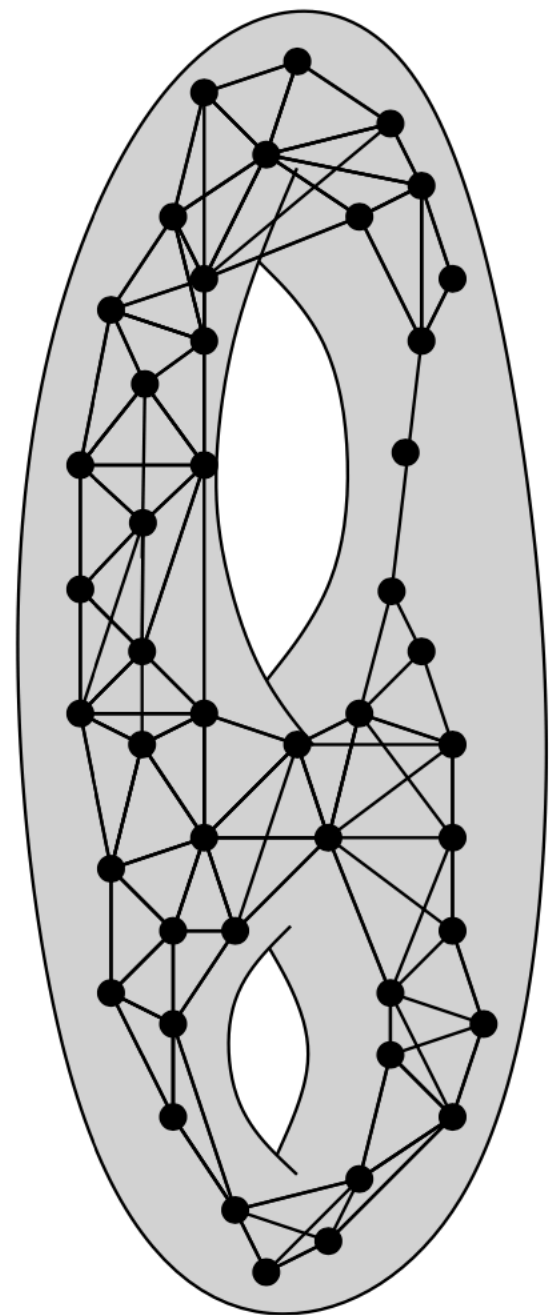
Let $\{C_i\}_{i \in I}$ be an open cover of a paracompact space X such that the intersection of every subcollection of C_i 's is contractible or empty.

Then, X and the nerve $N(\mathcal{C})$ are homotopy equivalent.

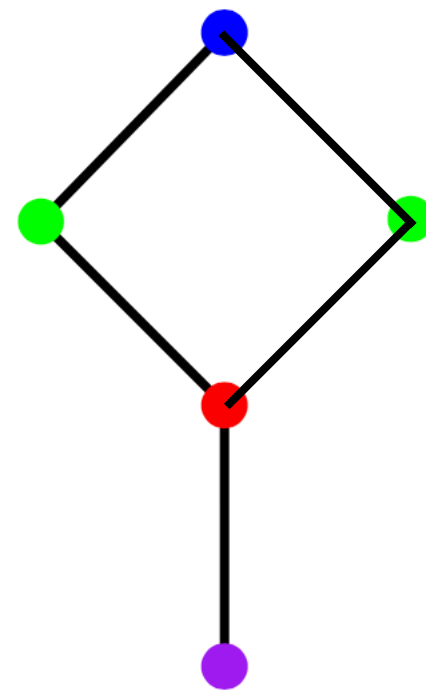
$N(\mathcal{C})$



Topological dataset summarization



X



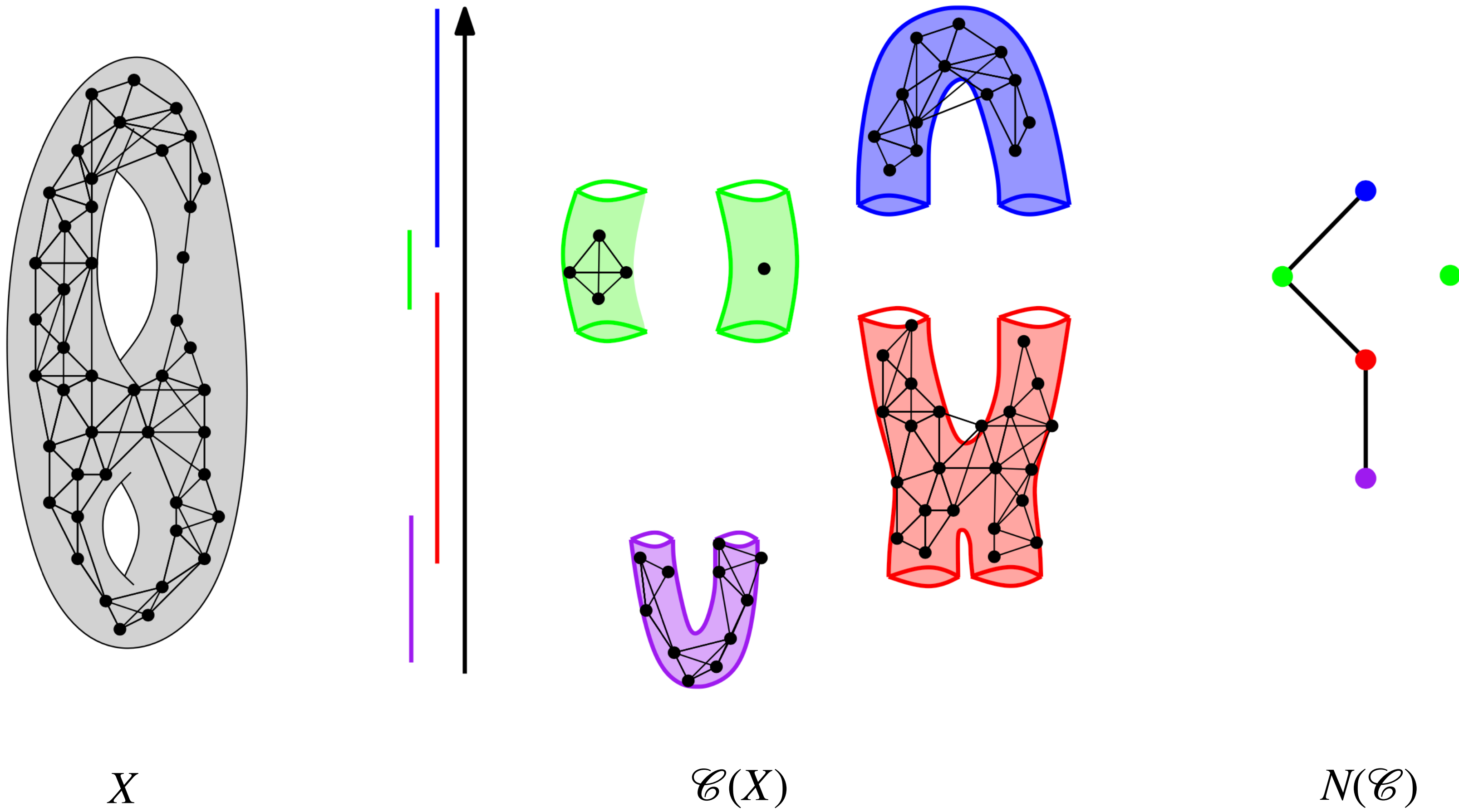
$K(X)$

Given a dataset X , topological dataset summarization seeks for a coarse topological representation of X in the form of a simplicial complex $K(X)$, that is

- the vertex set V of $K(X)$ is much smaller than X , $|V| \ll |X|$,
- the topology of $K(X)$ “resembles” the topology of X .

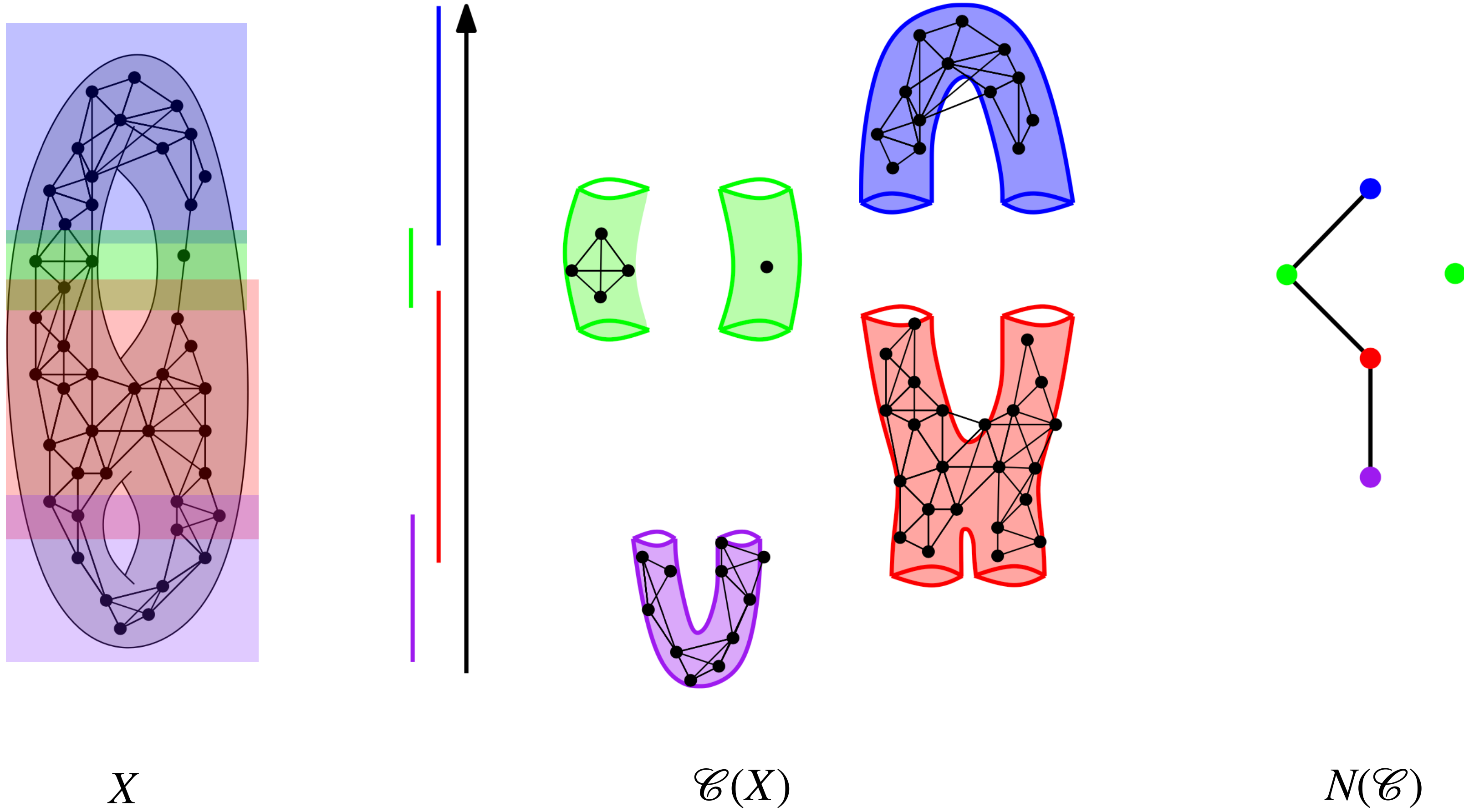
Topological dataset summarization

Nerve complex



Topological dataset summarization

Nerve complex



Reeb graph

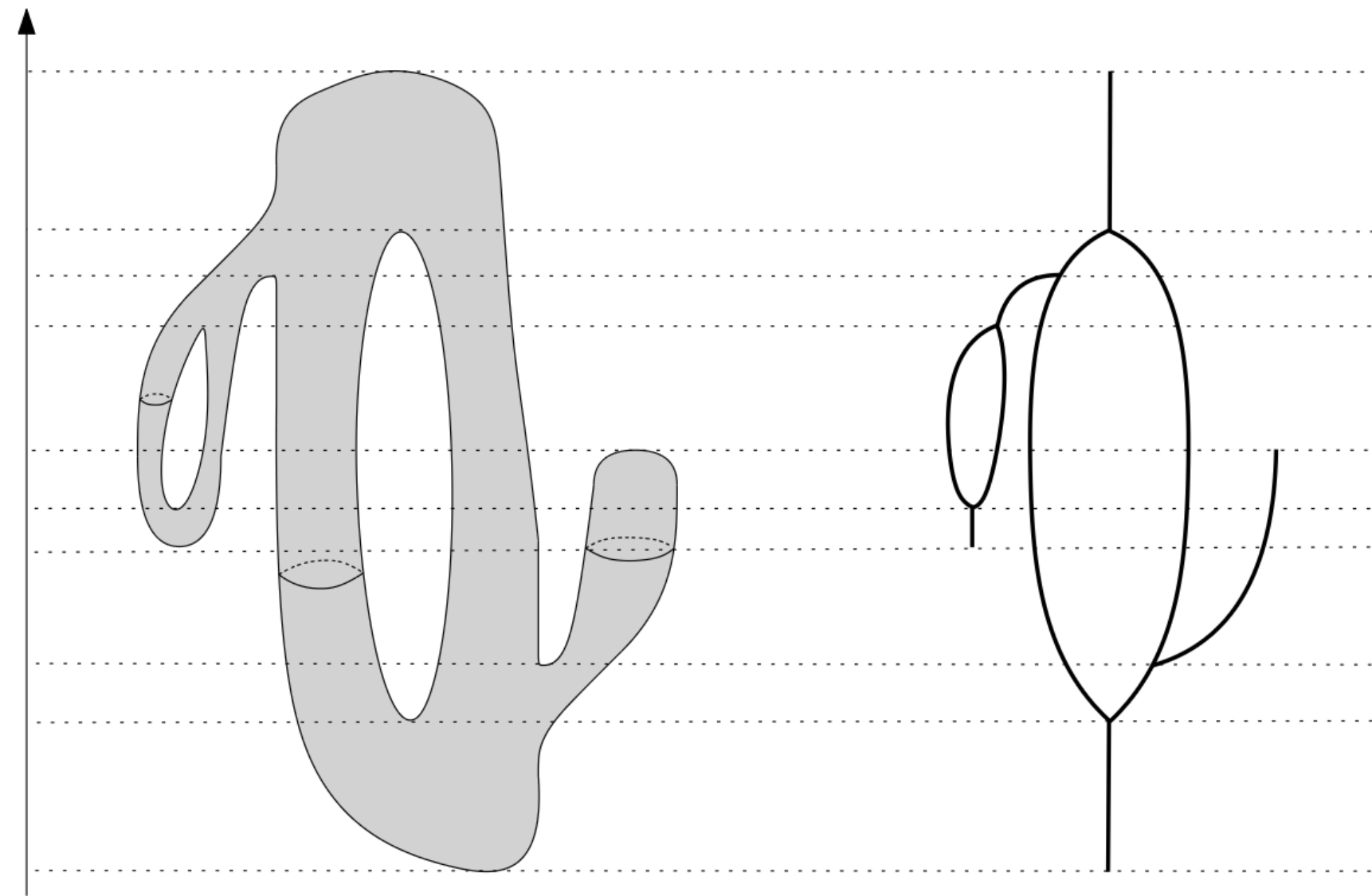
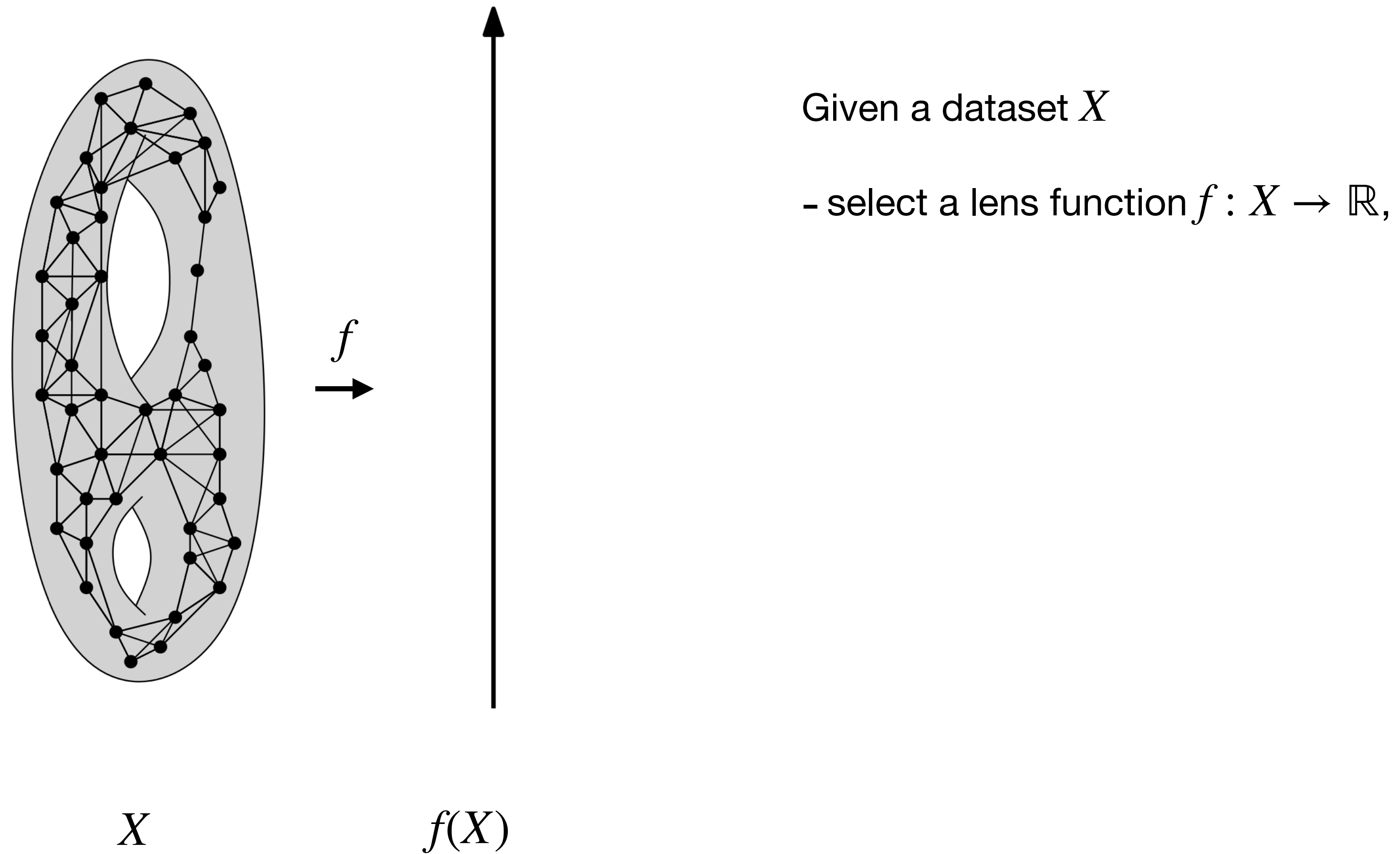
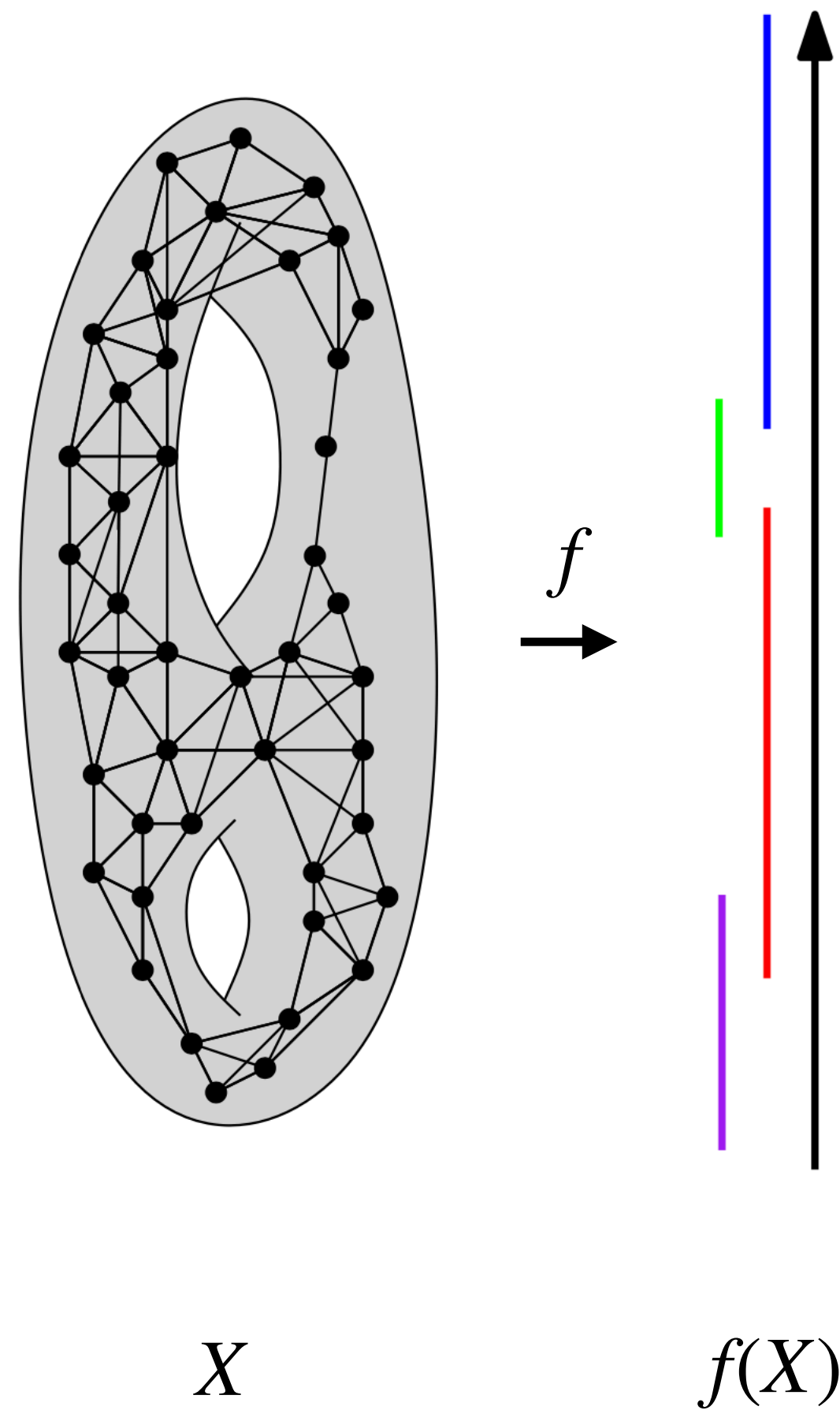


Figure 2: Example of Reeb graph computed on a double torus with the height function. Connected components of the level sets of the function (such as the three different ones drawn on the double torus) are contracted into single points.

Mapper



Mapper

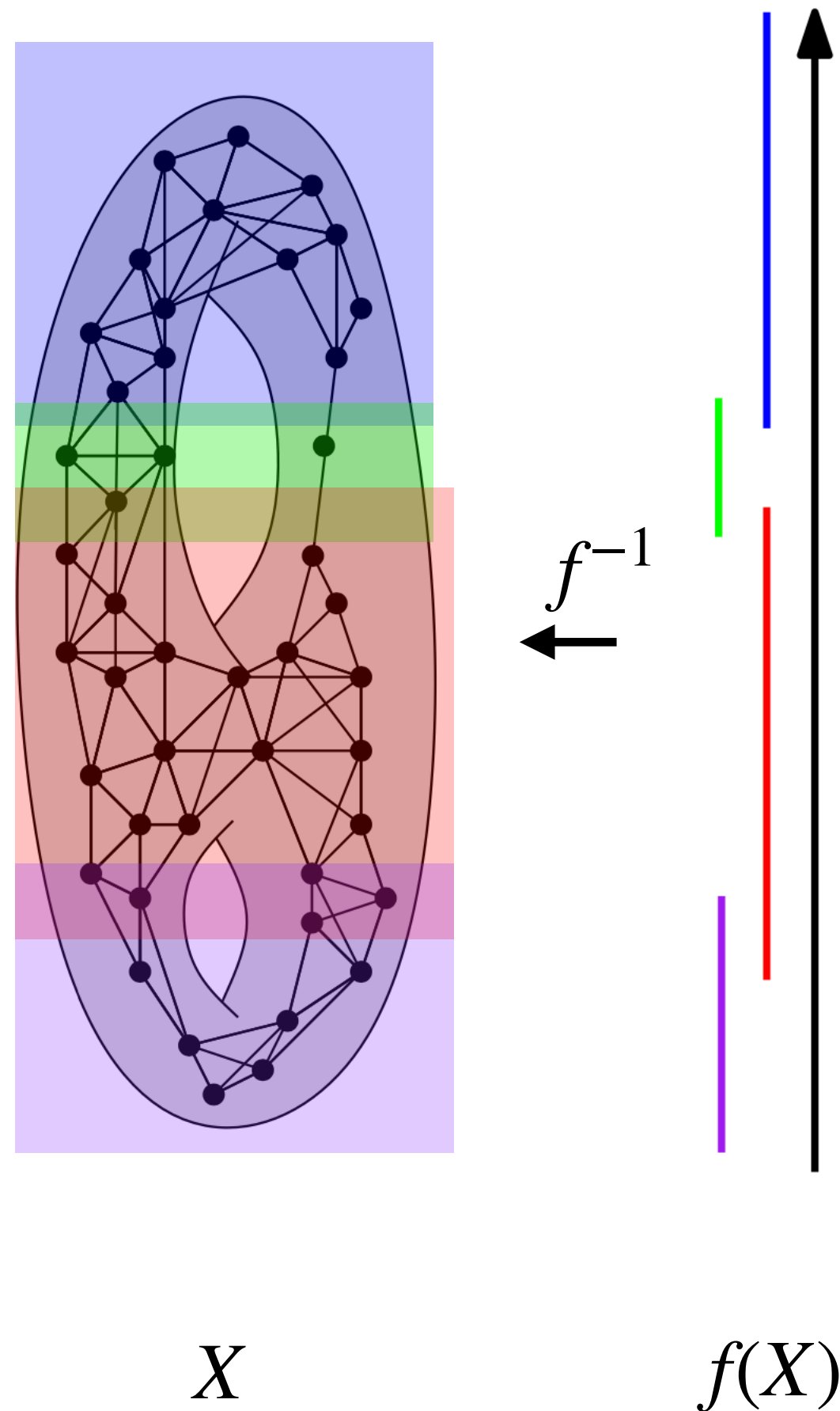


Given a dataset X

- select a lens function $f : X \rightarrow \mathbb{R}$,

- cover the range of the of f with overlapping intervals $\mathcal{C} = \{C_i\}_{1 \leq i \leq N}$,

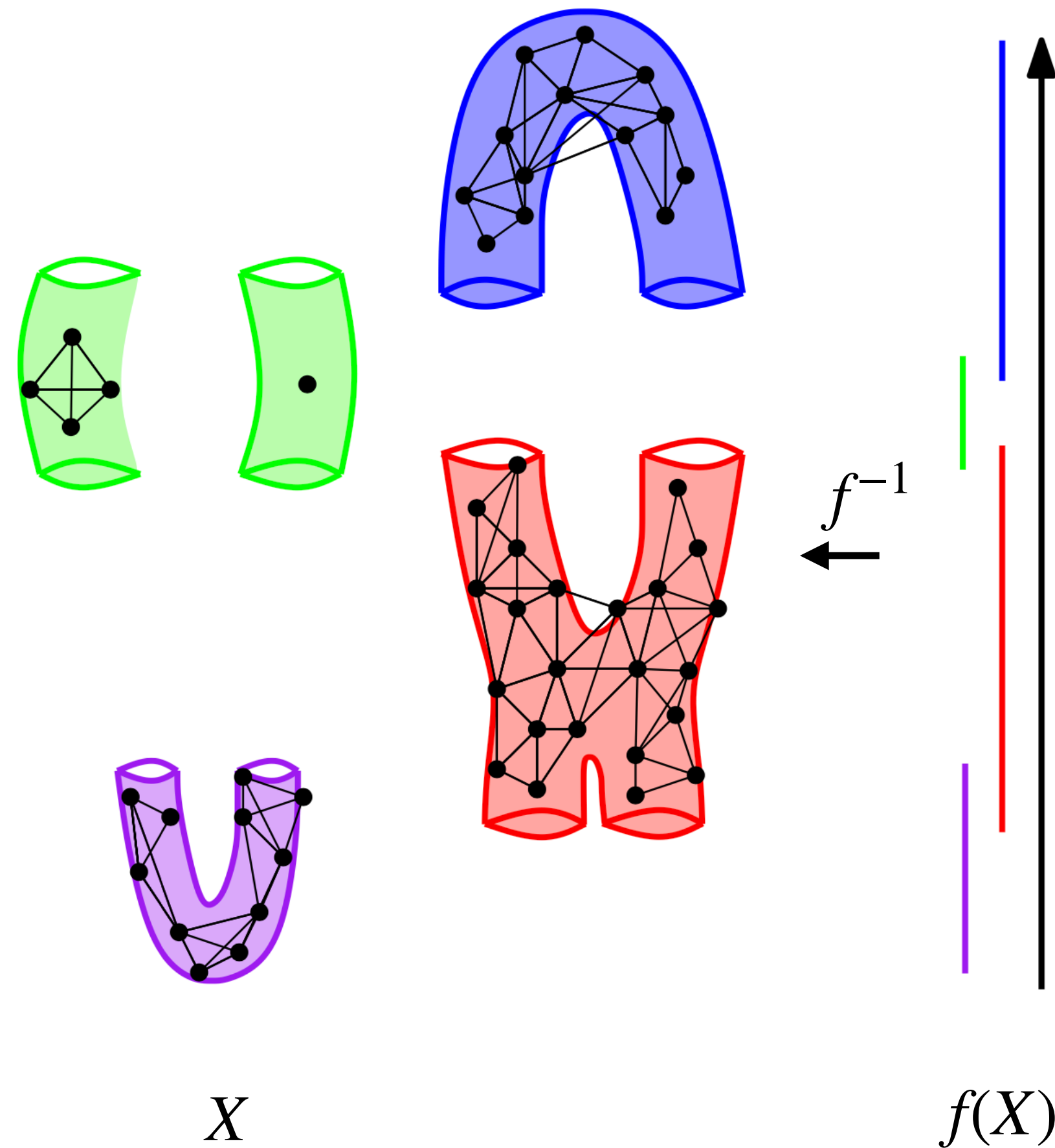
Mapper



Given a dataset X

- select a lens function $f : X \rightarrow \mathbb{R}$,
- cover the range of the of f with overlapping intervals $\mathcal{C} = \{C_i\}_{1 \leq i \leq N}$,
- find the pullback cover $\mathcal{C}^{-1} = \{C_1^1, \dots, C_1^{k_1}, \dots, C_N^1, \dots, C_N^{k_N}\}$ by a clustering algorithm, where C_i^k denotes the k -th cluster of $f^{-1}(C_i)$,

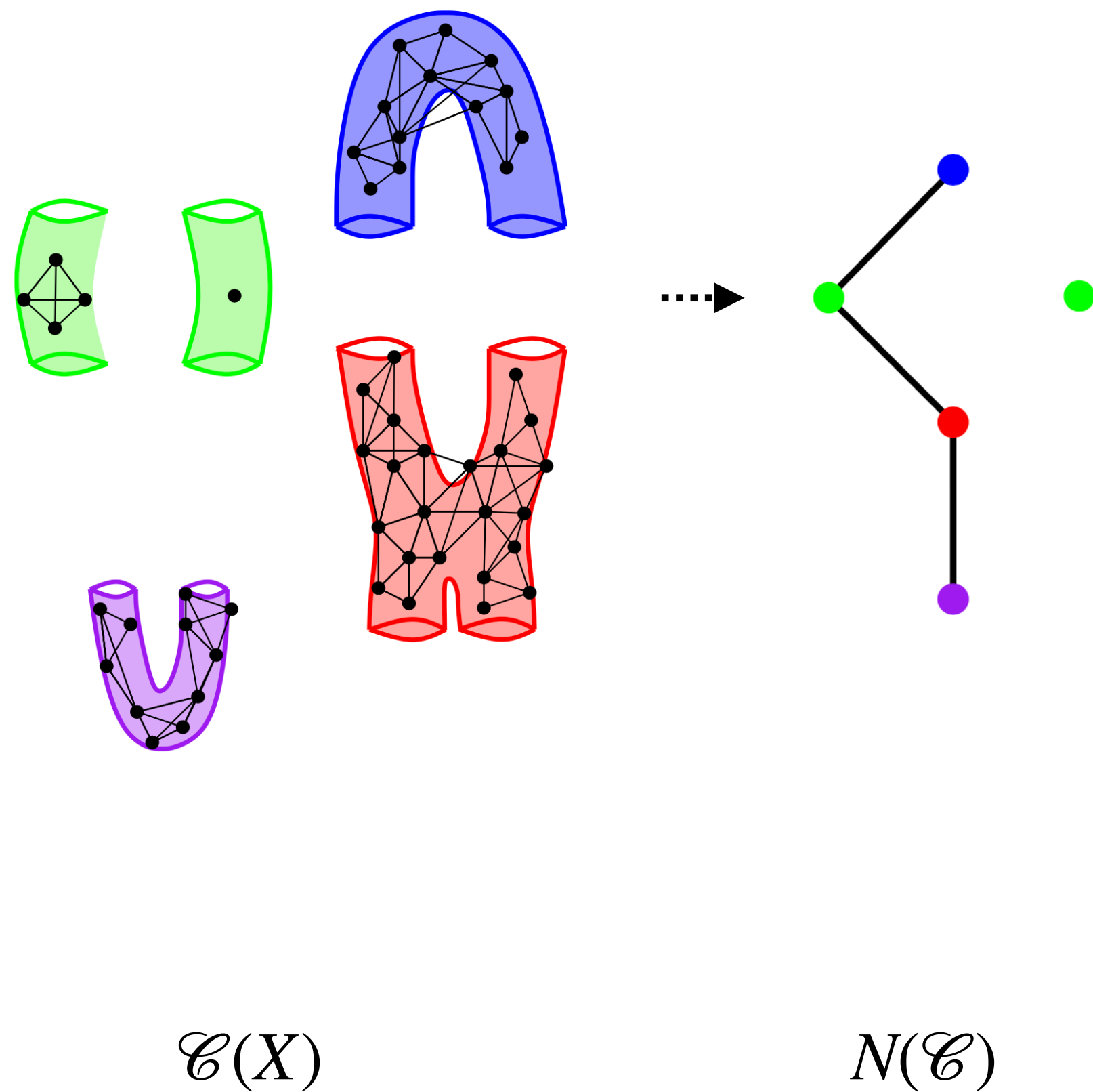
Mapper



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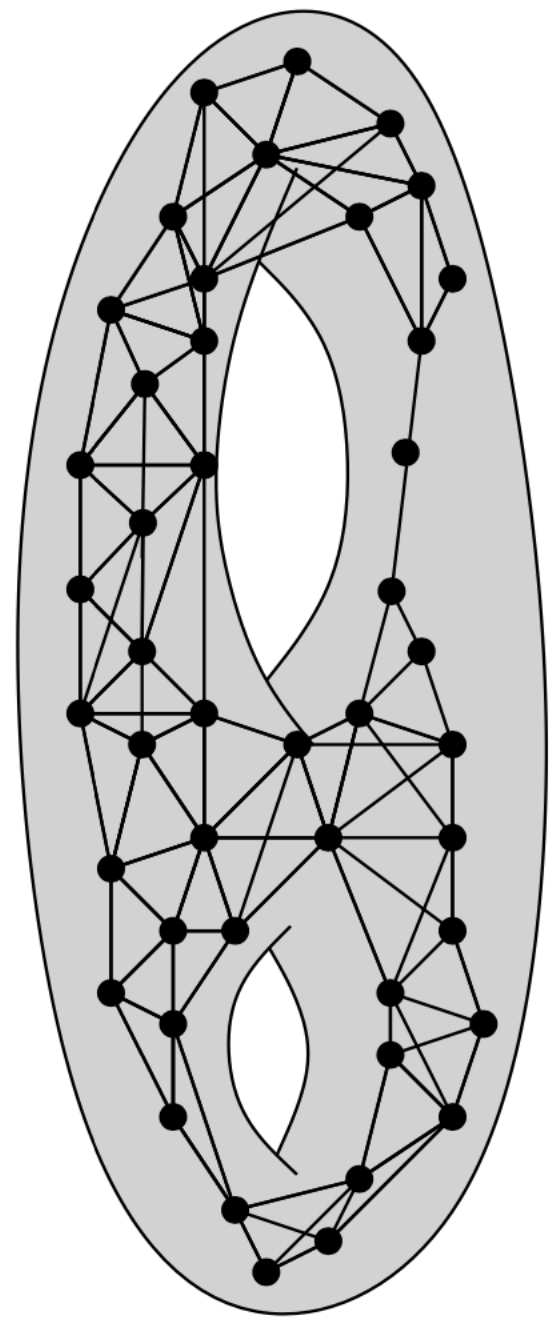
Mapper



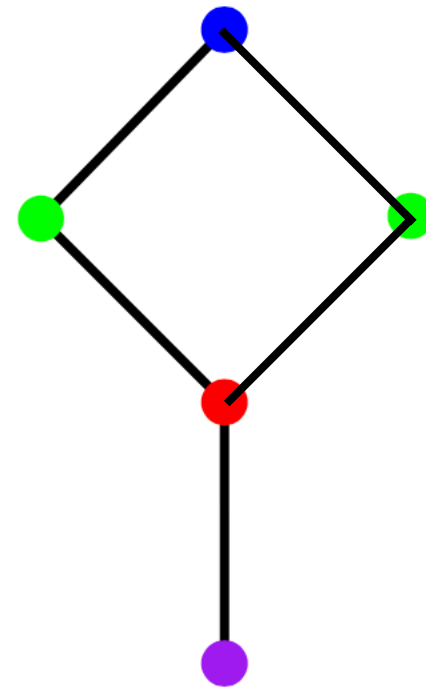
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- $N(\mathcal{C}^{-1})$, the Mapper of X is the nerve of a pullback cover $\mathcal{C}^{-1}$, that is vertex $v_i^k \in N(\mathcal{C})$ corresponds to C_i^k , and there is an edge $\{v_i^k, v_{i'}^{k'}\}$ iff $C_i^k \cap C_{i'}^{k'} \neq \emptyset$.

Mapper variants



X



$K(X)$

Choices

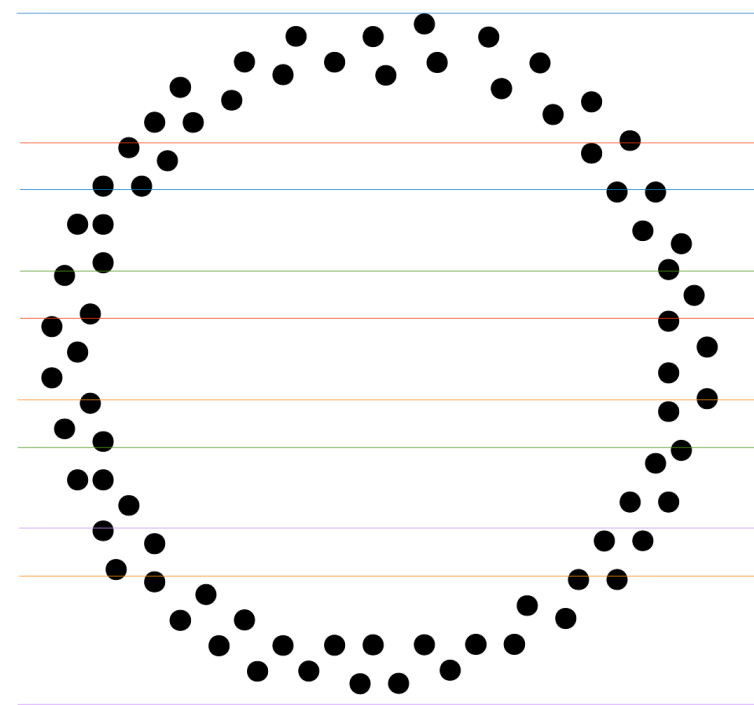
- a lens function $f : X \rightarrow \mathbb{R}$, more generally $f : X \rightarrow \mathbb{R}^n$
- way to construct cover $\mathcal{C}$ of f ,
- clustering algorithm to construct the pullback cover $\mathcal{C}^{-1}$.

Mapper

Lens functions

- coordinate projections
interesting features of data (survival rate for clinical datasets, time for time-dependent data, etc.)
- projections to first-n Laplacian eigenvectors
- probabilistic
density
- complexity
Kolmogorov complexity of a string
- geometric
intrinsic dimension, curvature

Ball Mapper



X

Given a dataset X and a proximity parameter $\varepsilon > 0$, construct an ε -net of X , that is a subset $L \subset X$ such that

$$\exists \ell \in L, d(x, \ell) \leq \varepsilon \quad \forall x \in X$$

Data: Point cloud X , $\epsilon > 0$

Result: $L \subset X$, an ϵ -net

$L = \emptyset$;

while *There exists a point $x \in X$ farther than ϵ from every point in L* **do**

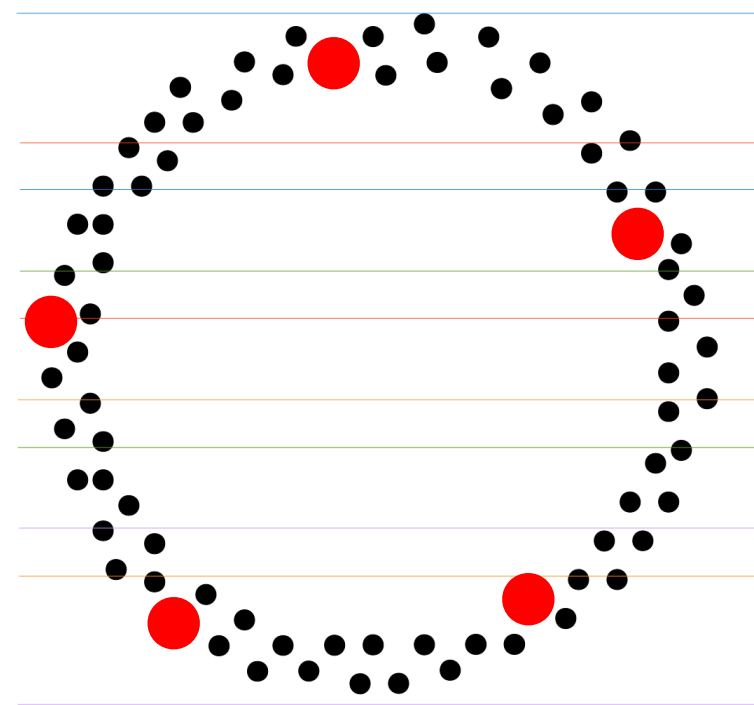
 | $L = L \cup x$;

end

return L

Algorithm 1: Greedy ϵ -net [33].

Ball Mapper



X

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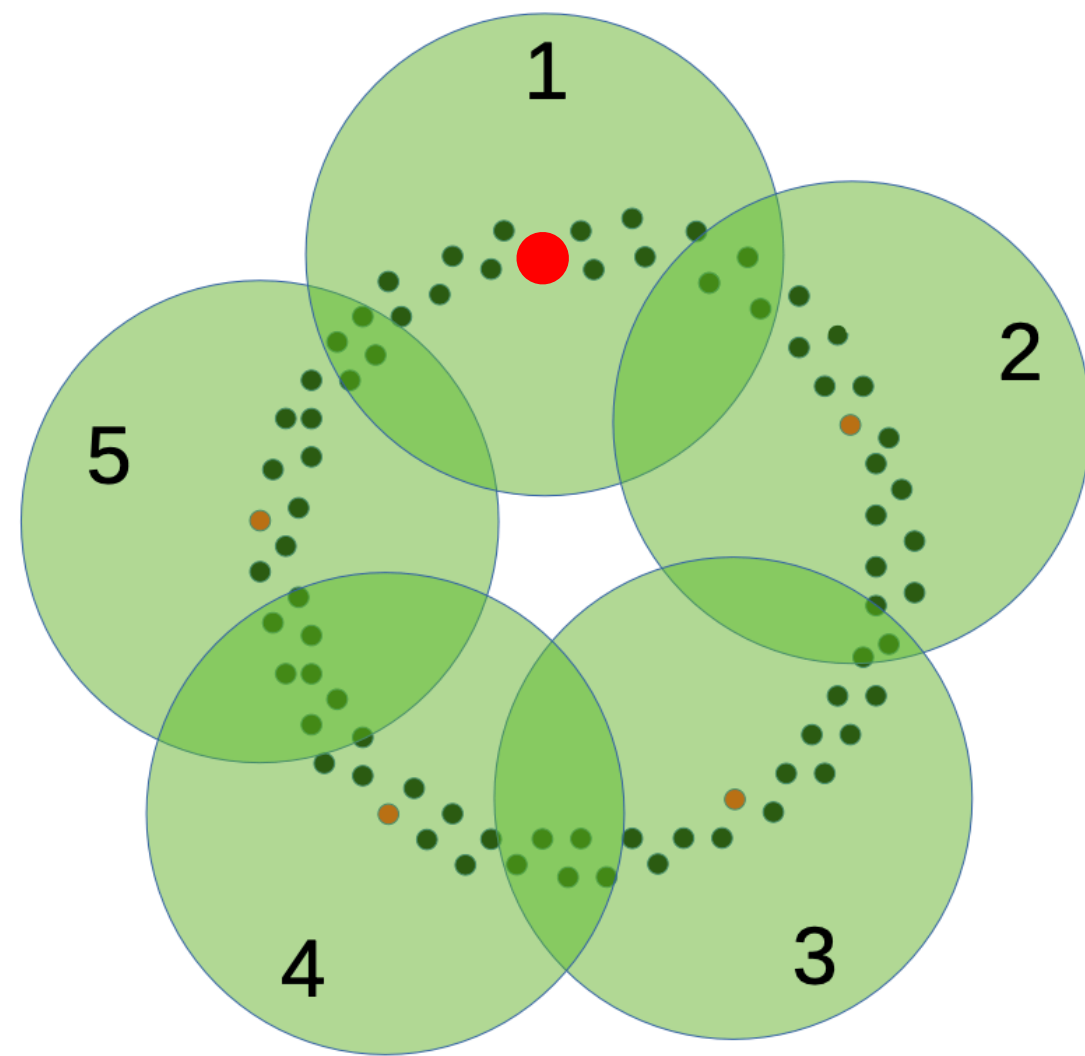
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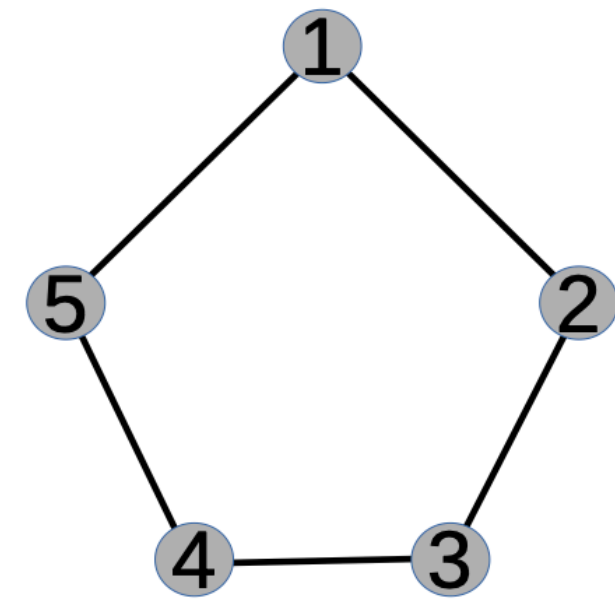
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Algorithm 1: Greedy ϵ -net [33].

Ball Mapper



$\mathcal{C}(X)$



$N(\mathcal{C})$

Given a dataset X and a proximity parameter $\varepsilon > 0$, construct an ε -net of X , that is a subset $L \subset X$ such that

$$\exists \ell \in L, d(x, \ell) \leq \varepsilon \quad \forall x \in X$$

Ball Mapper of X is the nerve of a union of balls $\bigcap_{\ell \in L} B_\varepsilon(\ell)$, that is

vertex $v_i \in N(\mathcal{C})$ corresponds to ℓ_i , and there is an simplex $\{v_i, v_{i'}\}$ iff

$$\sigma^{(k)}\{v_0, v_1, \dots, v_k\} \in N(\mathcal{C}) \iff B_\varepsilon(\ell_0) \cap B_\varepsilon(\ell_1) \cap \dots \cap B_\varepsilon(\ell_k) \neq \emptyset$$

D-mapper

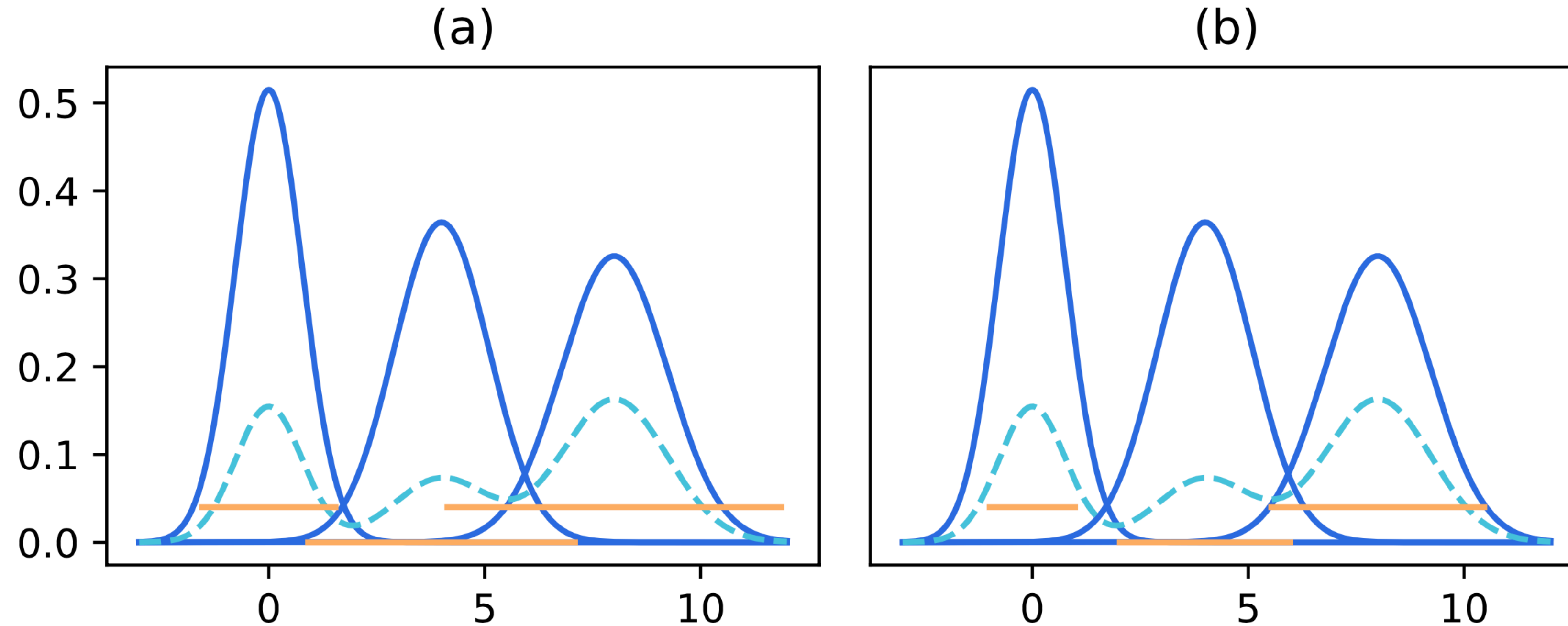
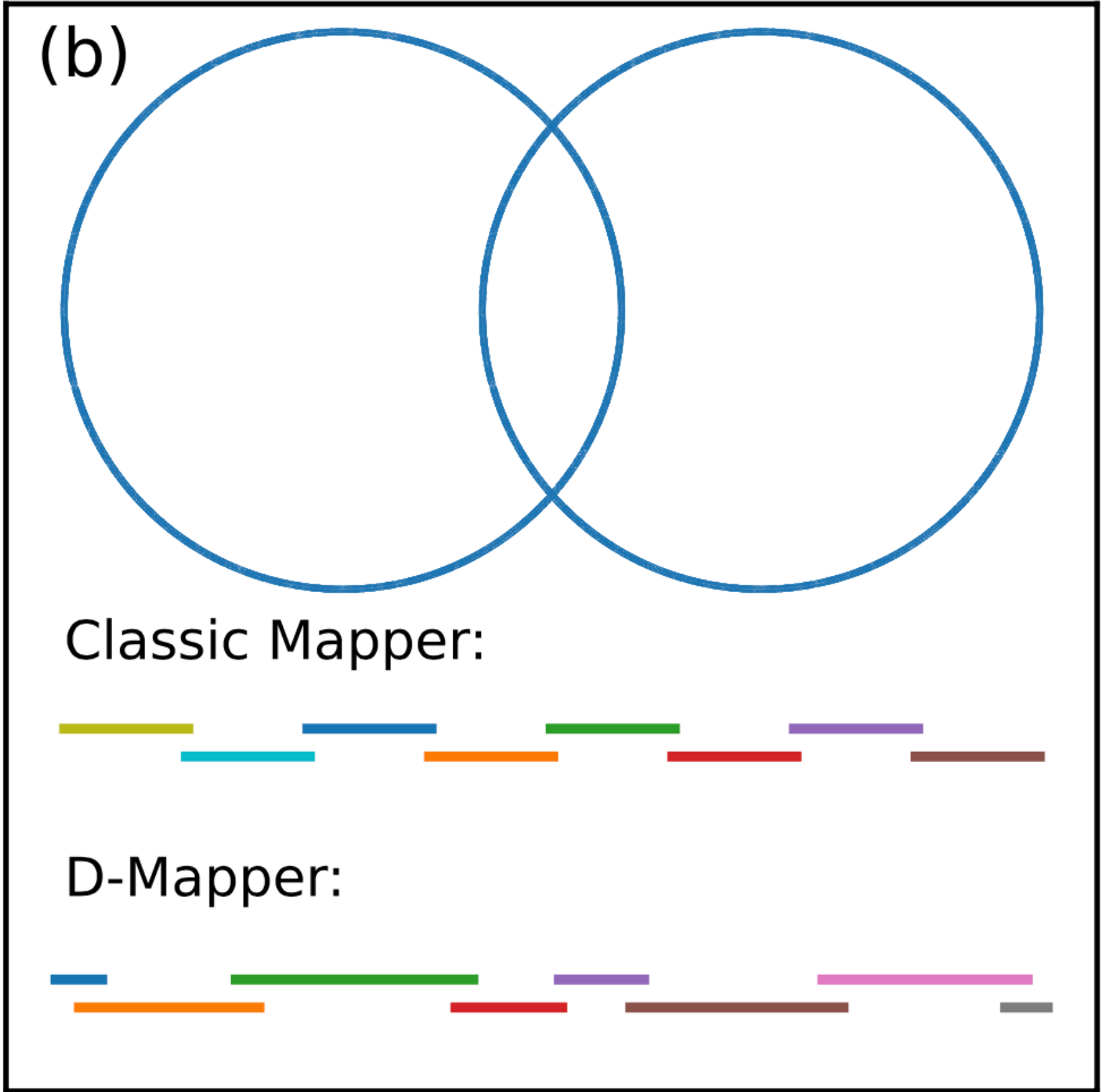
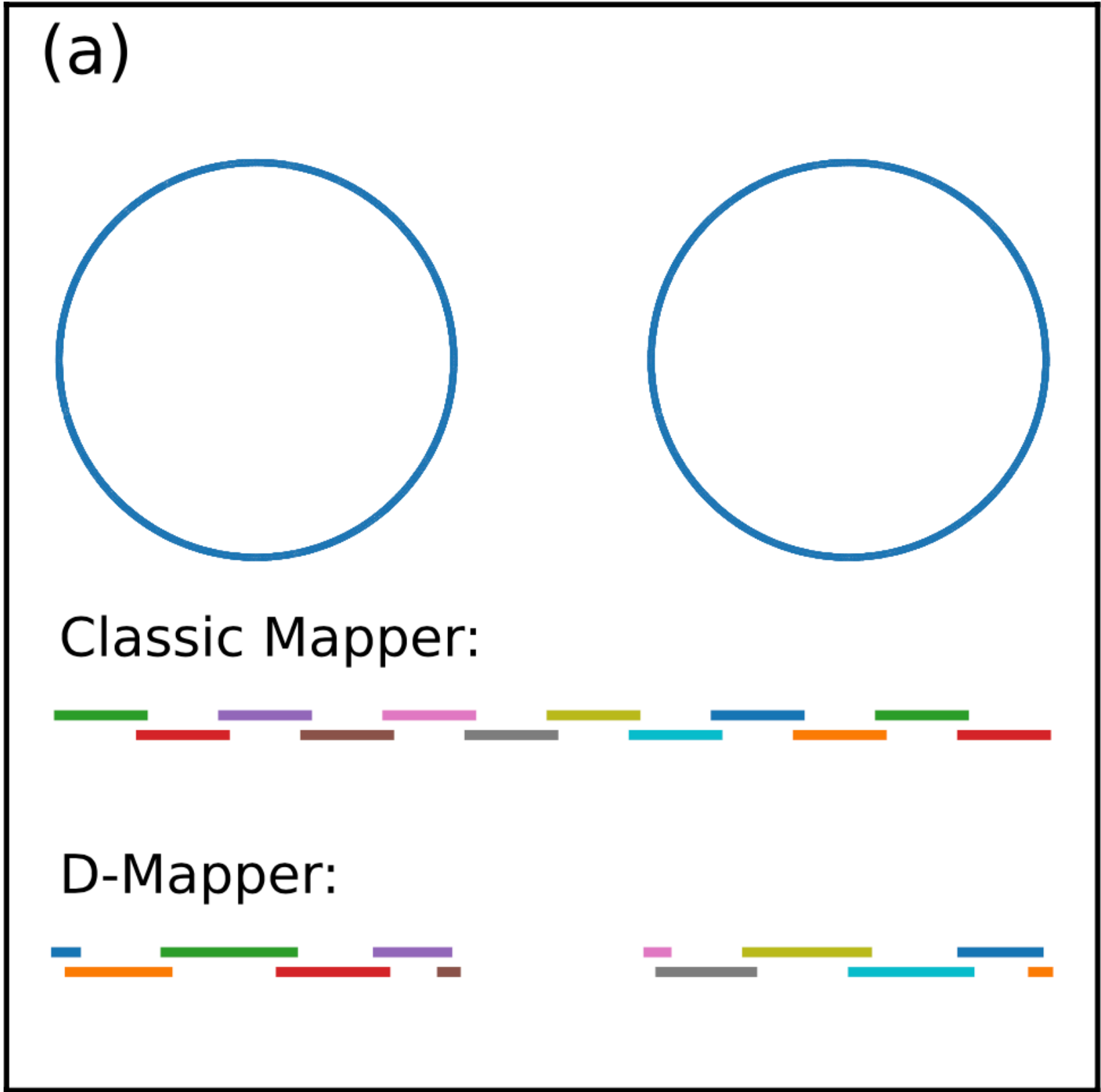
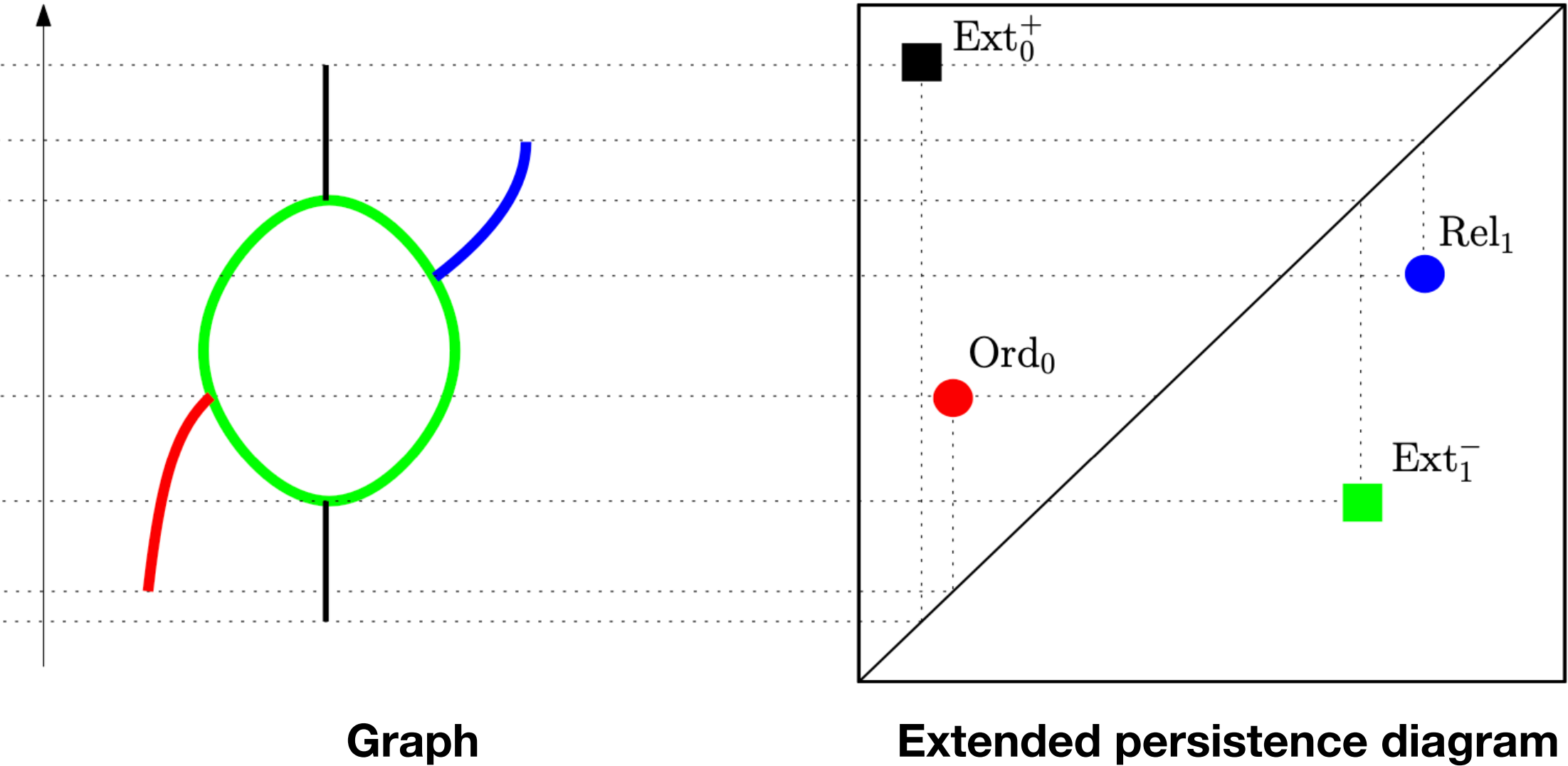


Fig. 1 An illustration of intervals produced by the D-Mapper algorithm. The deep blue line represents the probability density function of each component in the GMM. The shallow blue dashed line presents the probability density function of the GMM. The orange lines are intervals that are produced naturally given a confidence level of $1 - \alpha$. (a) When $\alpha = 0.01$, there are overlaps between adjacent intervals. (b) When $\alpha = 0.1$, there is a gap between the first and second intervals. α controls the overlap of intervals, and it should be chosen carefully.

D-mapper



Comparing Mappers



Distance between Mapper graphs as the distance between their extended persistence diagrams

$$d(G, G') = W_p(D^+(G), D^+(G'))$$

$$\emptyset = K_{-\infty} \subset K_1 \subset K_2 \subset \dots \subset K = (K, K^\infty) \subset \dots \subset (K, K^2) \subset (K, K^1) \subset (K, K^{-\infty}) = \emptyset$$



$$K_t = \{\sigma \in K \mid f(\sigma) \leq t\}$$

sublevel set $t \in (-\infty, +\infty)$

$$K^t = \{\sigma \in K \mid f(\sigma) \geq t\}$$

superlevel set $t \in (+\infty, -\infty)$